Supplement S1: Reference-Preserving Speculative Decoding on Consumer Systems

Author information pending confirmation

Abstract

This supplement records the output contracts, implementation boundaries, campaign-level results, negative findings, historical measurements and missing observations supporting the main manuscript. It preserves distinct task panels and timer conventions. The accompanying S2 author source package reproduces figures and arithmetic from committed exports without model inference. Verification here is limited to inspected sources and exports; inherited original-host raw audits are identified separately.

S1. Study Units, Contracts, and Frozen Sources

The main manuscript is organized by diagnosis, repair scope and cost, rather than by experiment date. Table S1 distinguishes the units used in the principal campaigns. S2's complete experiment-destination ledger assigns every inherited group to the main text, this supplement, or a preserved historical record, with exact repository entries. Discovery, engineering qualification, formal output checks, auxiliary crosses and capture-off costs are different evidence classes. Their counts are not added to create an independent sample size.

The late RTX target/draft pair is Qwen3-8B/0.6B Q8_0, with full CUDA offload, flash attention off, one slot, context 4096, caller batch capacity 2048, 24 threads and greedy seed 123. The ordinary reference is mask 0/UB1. Matched target-only and the anchor use mask 3/UB4, with the anchor's $k = 3$. The expansion starts from a frozen private server and CUDA libraries. Four separate resident processes per round each receive one excluded warmup and 24 formal requests. Condition order and task order rotate. Zero actual cache reuse is recorded; the final-slot `tokens_cached` field is not a cache-hit measure.

Table S1: Campaign units and distinct roles. Panels overlap unless described as project-unused at their own freeze.

Campaign	Units	Role
RTX continuation intervention	43 tasks, three repeats, two recipes	Predictions/native outputs; repairs and regressions
RTX natural crosses	Four probes; 24 donors, 48 forks, 24 shams	Selected aligned boundaries; auxiliary costs excluded
B570 single-block intervention	24 tasks, 72 runs, 195 events	Immediate and downstream effects
RTX phase factorial	16 masks, four probes, two repeats	128 qualification requests
RTX scope cost	19 exposed tasks; three conditions, six repeats	342 formal requests, 18 excluded warmups
RTX input expansion	24 project-unused MBPP tasks; four conditions, four repeats	384 formal requests, 16 excluded warmups
RTX fixed-UB width	28 exposed tasks; three widths, two repeats	168 qualification requests; separate 448-request cost arm
B570 natural crosses	Three tasks, six repeated contexts, 12 shams	State × whole recipe; 24 auxiliary cells
B570 expanded cost	15 project-unused tasks, four conditions, two rounds	119/120 payloads; 15 or 14 complete pairs
M4 formal runtime	60 tasks, three conditions, three repeats	540 formal requests; direct-ID gate
358H closed Phase 3	30 tasks, three repeats per draft device	90 direct-ID pairs per path

The 24 expansion inputs are unused in the accessible experiment inventory at selection. An unused HumanEval pool is unavailable at that stage; the amendment selecting MBPP is recorded before new outputs. Of 224 eligible MBPP inputs, 24 are frozen. This is project exposure accounting, not a

training-contamination claim or a random population sample. The width and post-hoc ordinary analyses subsequently reuse those exposed inputs.

RTX qualification compares every generated response ID and its EOG/cap128 termination metadata. B570 comparison selects the first inclusive EOG or cap128 endpoint from actual committed IDs; raw post-endpoint suffixes are retained. Historical parsed-text agreement re-tokenizes text and is weaker. M4 formal runtime and 358H Phase 3 use their own direct generated-ID contracts. A later strict-ID pass does not retrospectively qualify historical timings. Token equality is neither functional code correctness nor bitwise logit/KV equality.

The planning baseline is commit `00ea8f7c3a49b900bf5f21a95876e9a3a6dd610f`, a child of the v0.13.0 manuscript head. New v0.14.0 files preserve all inherited evidence and historical sources. S2 includes authenticated copies of selected original reports and implementation patches, alongside compact exports. Some full raw archives and external model/runtime assets are referenced rather than included. The CPU reconstruction in this revision never launches inference, warmups or model experiments.

S2. Diagnosis: Predictions, Interventions, and Natural Crosses

S2.1. Prediction closes a different question from reference recovery

The B570 target-partition predictor is frozen after 36/36 discovery comparisons. On 24 reserved HumanEval tasks, three repeats give 72/72 predictor/native complete endpoint agreements, while reference/native agreement is only 60/72. All four reference-divergent classifications and first-divergence indices reproduce. They are HumanEval/134 at index 88 (914 to 1946), /42 at 73 (432 to 1817), /145 at 103 (1184 to 686), and /138 at 93 (14374 to 9064). H1 passes its gate; H2 is not built in that B570 study. Stable predictive disagreement is not repair.

The RTX H1 transfer uses 12 discovery tasks with three repeats and reaches 24/36 against a 36/36 gate. It stops before any reserved inference; H2 is not implemented in that study. A later independently frozen proposal-replay study evaluates 24 reserved tasks. H2/native and a fresh draft-free R-only replay each reach 72/72; ordinary reference/native reaches 60/72. The replay explains complete native output under its numerical routines, but does not establish a cheaper qualified online correction. Its separately orchestrated workers share pinned routines with the native implementation; they are not an independent numerical implementation.

RTX V4 has 19 HumanEval tasks, with P/R and R-only/native each 57/57. Ordinary target-only UB2/native and UB1 reference/native each reach 51/57. The same task panel is used for fresh timing, not a newly untouched cohort. Section S6 records the host-policy amendment and preceding V3 termination separately.

S2.2. Controlled prefill and continuation interventions

B570 changes target prefill UB1 to UB128, while target continuation and draft stay UB1. The protocol also samples and accepts the first pending token and transfers KV, logits, logical positions and a post-accept sampler clone. Both baseline and intervention predictions match native execution 72/72. Reference agreement improves from 60/72 to 72/72: four exposed failures repair and 20 stable controls remain equal. This intervention does not isolate a pure prefill operator or KV-only cause.

RTX changes target continuation UB2 to UB1, leaving prefill, draft UB2, $k = 2$ and caller grouping fixed. Across 19 HumanEval and 24 statically selected MBPP tasks, both predictors match their native paths in all 129 comparisons per recipe. Reference agreement stays 117/129. HumanEval/15 repairs, /25 regresses and /18 remains divergent, giving 51/57 before and after. MBPP/304 and /463 remain divergent, giving 66/72 before and after. MBPP/304's 47-ID output changes at zero-based index 33 from 488 to 481. Aggregate equality of the success counts is therefore not a no-op result.

Unchanged-capacity shams test handoff at the same boundary. Historical global-UB1 also changed prefill/draft and effectively used 12 native threads; this controlled intervention explicitly uses 24. Its G0 qualification includes 75 newly checked native arrays at 24 threads and 45 unchanged worker arrays reused. It is not 120 new native 24-thread measurements. These differences prevent an argument that this intervention alone establishes universal phase necessity.

S2.3. Same-state single-block intervention on B570

The sampled-fork study uses the 24 reserved tasks, three repeats and a frozen boundary at the first eligible verification block after at least 64 committed IDs. Two paths start from the same retained donor state and process the registered block. Scores differ in all 72 runs, while immediate top-1 and decision outcomes remain equal across all 195 internal decision events. Downstream output changes in 18 runs from six tasks. The alternate continuation retains later UB128 blocks, with proposals evolving under its

accepted history; the secondary result does not isolate one altered block followed by restoration. Immediate agreement cannot stand in for complete trajectory agreement. Runs and decision events are not independent additional tasks; the raw verifier results cited here are inherited original-host results, not a new private-payload audit.

S2.4. Natural donor-state × current-recipe crosses

The RTX experiment uses four selected probes, three repetitions and two naturally formed donors at aligned committed boundaries. Each donor is forked under both current recipes, and own-path shams reproduce uninterrupted output. There are 24 donors, 48 forks and 24 shams. Target-only auxiliary replay is diagnostic rather than paid deployment performance.

For HumanEval/15, the shared committed cut is 13. At the third auxiliary decision, the two donor rows are identical: $[9147, 4199]$ under current UB2/UB1. Each full three-ID auxiliary array begins $[6256, 5987]$, followed by the displayed ID. At this cut, the observed decision follows current recipe. HumanEval/25 shares four committed IDs and its first-decision matrix is $[[1406, 1406], [1406, 271]]$. The bottom-right three-ID array is $[271, 750, 8168]$, whereas the other cells begin with 1406. This supports an interaction at the selected boundary. HumanEval/18 and /31 are local nulls under the measured crosses. Logical prompt KV matches in 12/12 donor comparisons, while generated-region KV differs in 70/72 layer-side comparisons; the first layer matches. Tensor inequality alone does not identify a unique kernel. The 1,278 recorded auxiliary forwards are not latency measurements.

B570's later natural crosses use six repeated contexts over three tasks, with 12 own-path shams. HumanEval/145 gives $[[71465, 279], [279, 279]]$ and /42 gives $[[432, 1817], [1817, 1817]]$: both support local state–recipe interaction. MBPP/72 gives $[[320, 320], [308, 308]]$, supporting a state effect; its current columns are deliberately identical (UB1, reservation 3). The two HumanEval recipes change both microbatch and output reservation, so this is a whole-recipe cross rather than an isolated UB factor. Recorded target score-array difference counts are 3,987,448, 5,218,133 and 6,072,820 across the three cases. Opaque sampler/RNG bytes are not independently available; the greedy contract, clone history and shams bound the claim. The later crosses do not retroactively become experiments in the earlier B570 closeout.

S2.5. Bridge and offline selection limits

The ordinary target-only U2 bridge compares 19 exposed tasks, three recipes and three repetitions: 171 fresh arrays. U2 reproduces intervention N1 at 57/57 complete arrays and strict boundaries, while both match ordinary reference on 17/19 tasks. U2 uses one occupied continuation token despite UB2 capacity. This simpler reproducer limits claims that the complex replay frontend is necessary; equal output does not imply equal numerical state.

The offline selector feasibility study compares existing evidence acquisition and decision rules. RTX direct enumeration and proxy acquisition each require 114 equivalents and abstain under the measured criterion. B570 direct enumeration uses 72 while the proxy uses 144. These equivalents are not measured latency. Unobserved MBPP U2 counterfactuals remain unknown. This study delivers a closed negative feasibility result, not a validated automatic online selector or an unfinished promise of one.

S3. Implementation and Phase Qualification

S3.1. RTX query layout and dispatch

The source-bound implementation modifies the non-flash attention graph, not the quantized projection/FFN kernels or the greedy acceptance algorithm. After the standard query permutation, a query tensor with dimensions $(d, q, h, 1)$ is viewed as $(d, 1, h, q)$. KV tensors broadcast over the query batch dimension. The corresponding mask columns are materialized with the matching layout; existing matrix multiplication, softmax, value multiplication and recombination operations then evaluate each query column with the registered geometry. Batched projection, FFN and output computation remain available. This shape description is about graph operations, not independently traced CUDA kernel dispatch.

Assertions restrict the tested graph to Qwen3 non-flash attention, one KV stream and supported query widths, without attention bias, sinks or MLA. The initial patch supports up to four queries; the width study extends the admitted graph to eight. Unsupported cases do not inherit qualification. The earlier implementation creates per-query attention subgraphs before the later query-batch transformation; their costs differ.

The four-switch patch records model role and phase through a thread-local guard entered before graph capture/fast-path selection. Target prefill and continuation have bits 1 and 2; draft prefill and continuation have bits 4 and 8. The process mask selects enabled phases. A disabled phase returns to the

native graph. The graph parameters and reuse check include the query-batch flag, preventing reuse across differing constraint modes. Reset re-enters prefill. Thus the switches act on actual graph selection; they are not a retrospective label on timings. Capture-off bypasses observer metadata and forced witness synchronization. The optional continuation state-move helper is inactive in the frozen expansion because its activating environment variable is absent.

The exact scripts, native delta and reports are included under `source_records/benchmarks/rtx_consistency_contract_v1` and `rtx_consistency_phase2`. S2 records their Git blob and SHA-256 identities. The portable manuscript code does not apply patches or execute these runtimes.

S3.2. Factorial and scope-cost results

All 16 phase masks are run on HumanEval/15, /18, /25 and /31 twice. The task-level booleans in the original A1 result agree with the committed 16-row aggregate export. Passing masks are 3, 7, 11 and 15; all repeat stably. Masks contain target prefill and continuation when they pass. HumanEval/15 and /31 pass throughout this family; /25 needs at least one target bit; /18 needs both. Figure 3 of the main manuscript shows every mask and probe rather than only successful configurations.

Mask 3 is inclusion-minimal among passing masks in this fixed four-switch family: no strict subset passes. It is not a statement that every runtime must constrain precisely these phases, that each bit is necessary on every input, or that all possible constraints have been searched. Natural crosses identify local dependencies; the factorial qualifies implementations under a different complete-output gate.

The independent scope-cost arm uses 19 exposed tasks, three conditions and six balanced repeats: 342 formal requests and 18 excluded warmups, capture off. Mask 3 speculation versus matched target-only gives generation/request speed $1.278076\times$ /$1.112977\times$. Mask 15 versus mask 3 cost gives $1.003569\times$ /$1.025686\times$, with 12/19 generation and 17/19 request tasks faster. The extra generation interval is $[0.997739, 1.009456]$. Per-round generation ratios are 1.008646, 0.998091, 1.004328, 1.003691, 1.007860 and 1.006777. The predefined optimization gate additionally requires generation speed at least 1.03, request speed at least 1.00 and at least four of six faster rounds. It is not met because the aggregate generation gain is too small. Qualification does not imply a speed gain from fewer constraints.

S3.3. Implementation variants and paid-boundary changes

Table S2 retains the early local implementation comparisons. Their baselines and observer costs differ from the final frozen expansion. Request gains against global UB1 include a prefill/path change; they are not matched speculative increments. The strict final local comparison disables observers and matches target graph, UB4 and prefill, giving 1.256449× generation and 1.077451× request over 19 task medians, with 19/19 and 16/19 tasks faster. All observations are from that campaign, not paired with later costs.

Table S2: Earlier RTX implementation variants. Ratios are baseline/candidate under each stated campaign.

Candidate; comparison baseline	ID gate	Generation	Request
Global serial $k = 2$; global UB1 target	57/57	0.677	0.795
Per-query $k = 2$; global UB1 target	57/57	0.746	1.041
Per-query $k = 3$; new frontend global UB1	76/76	1.084	1.562
Query-batch $k = 3$; new frontend global UB1	76/76	1.242	1.852
Query-batch $k = 3$; strict matched target	76/76	1.256	1.077

The ordinary $k = 3$ path in the second campaign qualifies only 68/76, with /18 and /25 failing in every repeat. Its timing cannot become a lossless speed claim. Serial, per-query and query-batch comparisons preserve both successful and unsuccessful cost outcomes.

S4. RTX Expansion, Statistics, Width, and Reset

S4.1. Complete expansion and two timer boundaries

The expansion contains exactly 384 complete unique formal keys: 24 tasks, four conditions and four repeats. All four conditions qualify 96/96 full responses; ordinary reference is stable in all repeats. Lengths range from 11 to 128 IDs; 23 tasks stop at EOG and one at cap128. No extra full-stream suffix is present in the selected response records. This does not certify any unrecorded internal sampling stream.

The retained server generation field is `predicted_ms`: the interval from its prompt/first-sample boundary to the last generation update, with the source's minimum one-microsecond floor. It charges draft and target work within the interval, excludes initial prefill and first sampling, and may exclude final bookkeeping. Resident-request time is client wall time for complete

non-streaming response receipt and parsing. Model load, process startup, excluded warmups, boundary telemetry and shutdown are outside it; post-response asynchronous cleanup is not certified inside it. Neither timer is a component profile. Subtraction of generation from request does not identify a host, KV or attention cost.

For each condition/task, take the median of four absolute times, form baseline/candidate ratios, then their unweighted geometric mean across tasks. Table S3 preserves all 12 timer comparisons. The ordinary-reference comparisons are post-hoc CPU analyses of already paid, qualified observations. They are not new measurements or a new preregistered primary campaign. The mathematical identity $R/C = (R/M)(M/C)$ holds for 48 task-timer cells and both aggregate timers; it is not an additive cost decomposition.

Table S3: All RTX expansion comparisons, baseline/candidate. Each uses 24 task medians; C_3/C_{15} are mask 3/15 speculation.

Ratio	Timer	Speed	Descriptive interval	Faster
R/C_3	Gen.	1.170124	[1.117013,1.217681]	22/24
M/C_3	Gen.	1.169238	[1.117115,1.217153]	22/24
R/M	Gen.	1.000758	[0.995290,1.007752]	11/24
R/C_{15}	Gen.	1.152348	[1.099880,1.199192]	22/24
M/C_{15}	Gen.	1.151475	[1.099968,1.198554]	22/24
C_{15}/C_3	Gen.	1.015426	[1.007525,1.024253]	20/24
R/C_3	Request	1.741569	[1.654608,1.827079]	24/24
M/C_3	Request	1.035071	[1.004552,1.063481]	17/24
R/M	Request	1.682560	[1.589801,1.766329]	24/24
R/C_{15}	Request	1.690202	[1.610916,1.768226]	24/24
M/C_{15}	Request	1.004542	[0.975846,1.031662]	12/24
C_{15}/C_3	Request	1.030391	[1.022464,1.038939]	23/24

The primary descriptive CPU bootstrap uses NumPy Generator PCG64, seed 20261008, 10,000 task resamples of 24 indices in saved task order, and one shared index matrix for comparisons. Percentiles use linear/type-7 interpolation. The historical interval convention uses Python `random.Random.choices`, seed 20261005, 4,000 draws and sorted indices 99/3899. S2 retains and checks both conventions without replacing one with the other. No bootstrap treats repetitions or conditions as additional independent inputs.

Tables S4 and S5 give all absolute medians and primary ratios. The seven

matched-request regressions are listed explicitly beneath the latter. Four matched-request round geometric means are retained; round 2 is 0.940220×. Task resampling cannot remove that observed reversal or estimate independent day/session uncertainty.

Table S4: RTX generation absolute task medians (ms) and ratios. MBPP task IDs follow the saved order.

Task	R ms	M ms	C_3 ms	R/C_3	M/C_3
439	590.217	553.925	516.623	1.142453	1.072203
415	2870.787	2805.883	2469.301	1.162591	1.136307
124	408.370	416.421	495.281	0.824523	0.840778
234	276.411	279.562	243.149	1.136797	1.149754
441	347.456	350.851	258.048	1.346478	1.359637
229	1106.948	1104.535	823.986	1.343408	1.340479
12	458.673	453.639	368.023	1.246319	1.232641
458	273.815	275.552	243.791	1.123157	1.130278
299	454.850	457.803	386.201	1.177755	1.185404
443	1084.726	1079.483	898.341	1.207477	1.201642
18	588.585	588.574	535.248	1.099649	1.099629
93	253.047	253.966	200.438	1.262473	1.267058
437	296.111	295.709	248.526	1.191467	1.189851
230	341.034	344.760	255.338	1.335622	1.350213
253	478.293	476.294	367.254	1.302351	1.296908
436	431.820	432.591	325.377	1.327139	1.329507
115	431.379	433.787	402.353	1.072139	1.078124
133	475.426	478.179	421.431	1.128124	1.134655
448	1609.686	1615.810	1311.434	1.227425	1.232094
274	1023.130	1019.728	895.225	1.142875	1.139076
454	274.751	273.346	246.048	1.116658	1.110950
393	386.363	387.622	395.548	0.976780	0.979963
242	229.418	233.043	192.592	1.191210	1.210035
397	478.100	476.432	415.627	1.150309	1.146296

Table S5: RTX request absolute task medians (ms) and ratios. MBPP task IDs follow the saved order.

Task	R ms	M ms	C_3 ms	R/C_3	M/C_3
439	1337.685	804.939	825.507	1.620439	0.975084
415	3638.915	3056.087	2802.469	1.298468	1.090498
124	1087.418	650.070	779.296	1.395385	0.834176
234	971.130	521.843	519.011	1.871115	1.005455
441	1054.836	587.618	541.248	1.948898	1.085673
229	2389.621	1502.739	1305.581	1.830312	1.151012
12	1261.199	714.779	684.524	1.842448	1.044200
458	952.067	520.996	524.541	1.815049	0.993242
299	1190.416	693.830	659.236	1.805751	1.052475
443	1823.804	1324.677	1197.541	1.522958	1.106164
18	1392.036	851.751	824.704	1.687921	1.032796
93	1029.573	512.864	491.369	2.095314	1.043744
437	975.759	524.353	528.602	1.845923	0.991962
230	1254.875	639.720	590.859	2.123815	1.082695
253	1239.853	730.717	654.826	1.893408	1.115894
436	1132.163	660.519	597.764	1.893997	1.104983
115	1189.749	684.664	697.826	1.704936	0.981139
133	1270.639	732.764	714.316	1.778819	1.025826
448	2319.676	1846.448	1578.762	1.469300	1.169555
274	1876.304	1296.957	1220.322	1.537548	1.062799
454	953.416	509.887	523.471	1.821334	0.974050
393	1064.315	627.877	666.169	1.597667	0.942519
242	954.557	478.594	466.815	2.044832	1.025233
397	1163.200	708.823	701.044	1.659239	1.011096

Matched-request regressions: MBPP/439, MBPP/124, MBPP/458, MBPP/437, MBPP/115, MBPP/454, MBPP/393. Round GMs: 1.028699, 0.940220, 1.051749, 1.044236.

S4.2. Geometry failures and their natural-state check

The wider-geometry boundary arm uses 11 exposed tasks, (k, UB) pairs $(1,2),(3,4),(5,6),(7,8)$, caps 128/256/512, three modes and two repetitions: 792 formal requests and 24 excluded warmups. MBPP/229 fails at $k = 5$/UB6 and $k = 7$/UB8 across all caps and repeats, 12 failures. Ordinary and matched wider target-only controls pass this task. The earlier outside-panel

HumanEval/18 wide failure remains a separate boundary; it is not erased by a panel rule. Maximum actual output length is 261, so cap512 is not evidence of 512 generated positions on every task.

At a shared 47-ID prefix for MBPP/229, naturally formed ordinary and constrained donors are crossed with current UB6 layouts. Four own-path shams over two caps pass. The donor state changes whether EOG is selected under both current layouts; current geometry changes scores without independently flipping this observed decision. EOG-minus-alternative margins are $[-0.09283, -0.62923]$ for one donor and $[0.99707, 1.48868]$ for the other. They are selected-alternative margins, not true top-two margins. Maximum measured score differences are approximately 0.44920 and 0.38070. This is a local same-prefix witness, not a complete counterfactual suffix, a unique kernel identification or a new independent task for each cap.

S4.3. Fixed UB4 and width costs

Holding mask 3 and UB4 fixed, $k = 3, 5, 7$ each qualifies 56/56 requests on 28 exposed tasks in two repeats. The widest passing $k = 7$ is selected from qualification alone, followed by a separate capture-off cost arm with four conditions, four repetitions, 448 requests and 16 excluded warmups. Table S6 retains pooled and saved MBPP/HumanEval partitions. The ordinary wider speculative condition fails qualification and has no admissible speed result.

Table S6: Fixed-UB4 width costs: baseline/candidate. Anchor is $k = 3$, wide is $k = 7$, and matched has no draft. Partitions are descriptive.

Panel	Ratio	Generation	Request
All (28)	Anchor/Wide	0.829442	0.889868
All (28)	Matched/Wide	0.997404	0.950353
All (28)	Matched/Anchor	1.202501	1.067970
MBPP (24)	Anchor/Wide	0.805961	0.872693
MBPP (24)	Matched/Wide	0.953392	0.918750
HE (4)	Anchor/Wide	0.985408	1.000258
HE (4)	Matched/Wide	1.307590	1.164148

Formal draft/accepted counts are 3,528/2,772 at $k = 3$ and 5,392/3,192 at $k = 7$. Separate captured qualification counts give 305/232/199 target caller calls for $k = 3/5/7$, but 305/463/397 micrographs. They include their own warmups and witness synchronization and are not a kernel-launch

profile or a denominator for the formal timing. Long-cap fixed-UB checks remain qualified, with matched speed about 1.0001 × /0.9649× at cap256 and 0.9715 × /0.9340× at cap512. The caps, timers and small exposed task panels remain distinct from the expansion.

S4.4. Resident reset as a finite serial check

The resident D study contains 100 reset/capture requests including four warmups, and a separate 96-request cost arm with four excluded warmups. Two task orders retain reference output for the anchor recipe. Matched generation/request speed is approximately 1.1812 × /1.0385×. It checks reset semantics in a serial resident loop. It does not establish concurrent scheduling, throughput, multi-slot behavior or qualification of the widened draft in a resident setting.

S5. B570 Repair, Missingness, Suffixes, and Paid Costs

S5.1. Earlier finite closeout: a positive local case and failures

The earlier B570 closeout's normative audit establishes completion of 1,456 core records (`COMPLETE_FINITE_SCOPE_AUDITED`). Historical report wording about incomplete normative closure is superseded by that completed audit, not treated as a still-open experiment. The original findings remain finite: on the shared 24-task MBPP panel, the repaired old N1 qualifies but gives only 0.188525 × /0.179055× generation/request speed against ordinary R, and 0.804023 × /0.693541× against matched target-only. The old N0/N1 comparison is 0.999522 × /1.247059×: request progress against N0 does not imply ordinary-reference benefit.

On eight project-unused MBPP inputs, N1 and its matched path fail MBPP/72 in all four repeats. Five UB2 alternatives retain failures involving HumanEval/42, /145 or /138 and receive a NO-GO decision. The local $k = 3$/UB4/cap128 candidate on seven exposed longer probes is a positive finite case, with 1.345798× generation and 1.048339× request speed. It does not replace the failed expanded qualification. Long-cap anchor qualification passes 84/84 at caps256/512, with longest actual response 261 IDs. No natural state cross occurs in this earlier study; the crosses in Section S2 are later experiments.

A wrongly placed UHD730 campaign of 190 requests is rejected as infrastructure evidence rather than scientific B570 data. Its records and recovery remain preserved. Recovery does not authorize outcome-dependent retuning

or replacement of failed tasks. The later phase-binding candidate is a distinct implementation.

S5.2. Same-context phase binding

The B570 candidate preserves one live target context and sampler. Ordinary-reference prefill uses UB128 with caller output reservation 512/per-sequence reservation 1; the frontend samples and accepts the pending first token once. It then binds continuation UB1 and synchronizes/reserves the scheduler before native $k = 2$, draft-UB1 proposal, acceptance, bonus and rollback. The private API patch changes phase parameters and reservation within the context; it adds no model arithmetic, Vulkan shader or precision change. It has no task-ID routing, known reference token, margin-based correction or extra reference forward.

Unlike old N1, it removes a second target context, state handoff and three serialized witness readbacks. Capture-on checks compare KV before and after binding; capture-off avoids witness readback. Those checks do not supply a separate profile of the saved cost. The frontend still pays for synchronization, phase reservation, draft work, target work and reconciliation. S2 includes `phase.h`, `target.cpp`, `native.cpp` and the private-library delta with source bindings.

S5.3. Qualification counts and unresolved raw comparisons

Table S7 retains endpoint and raw-array distinctions. The 48 baseline-development rows have only 32 ordinary-reference-equal endpoints; they are not all qualified candidate evidence. The 12 shams are separate from the 299 rows in the qualification export. Stage D contains 56 exposed tasks evaluated twice. Stage E combines eight bridge records and 119 formal records; their roles are not interchangeable.

The four-condition formal matrix is 119/120. The missing unit is `E/FORMAL/1/old_N1/MBPP/400`, with `INFRASTRUCTURE_INTERRUPTION_NO_PAYLOAD`. Its runtime cost and interrupted teardown time remain unknown. Each of ordinary R, matched M and candidate C has 30 complete formal observations and 15 complete two-round task pairs. Old N1 has 29 observations and only 14 complete task pairs. The available MBPP/400 old-N1 round is retained but cannot supply its two-round median. There is no interpolation, zero-cost assumption, silent task removal from other comparisons or scientific rerun. The study's administrative closeout remains `PARTIAL_INFRASTRUCTURE_BUDGET_CLOSED`.

Table S7: B570 candidate qualification stages and raw comparison limits. Equality is to the registered endpoint reference unless stated otherwise.

Stage	Candidate checks	Endpoint equal	Raw reference comparison
C development	12	12	Four equal, eight different
D exposed bridge	112	112	MBPP: 16 equal/48 different; 48 HumanEval raw comparisons unknown
E candidate formal	30	30	Included in the E total below; suffix cost retained
E all available plus bridge	127	127	80 equal, 47 different

S5.4. All measured timers and work

Generation time is the frontend's registered C++ interval, including pending-state initialization, sampler/batch operations, phase binding within that interval, draft/verification, rollback and final synchronization. Whole-request time additionally includes prefill/reset/cleanup, excluding subsequent disk file writes. Host request time is separately measured by Windows QPC. Earlier A/B stage GetTickCount's 15.625-ms granularity is not substituted for formal C/D/E QPC measurements. Model-load cost is outside these comparisons.

Table 4 of the main manuscript reports all three formal timer aggregates. Table S8 provides all paired task generation and whole-request ratios. R/C and M/C are below one on every complete task and round. Old-N1/C whole and host request ratios reverse on MBPP/245 in round 2, approximately 0.873582 and 0.873603; aggregate old-path improvement therefore does not imply universal improvement.

Table S8: All B570 two-round task-median speed ratios. G/W denote generation/whole request; old N1 has no complete pair for MBPP/400.

MBPP	R/C G	R/C W	M/C G	M/C W	N1/C G	N1/C W
125	0.320296	0.270051	0.319891	0.243047	2.437945	1.908679
130	0.299904	0.252594	0.304551	0.257891	2.269168	1.978931
132	0.264679	0.162455	0.271252	0.169635	8.515349	4.089033
245	0.298694	0.254655	0.298448	0.255268	1.541642	1.376354
278	0.270026	0.205714	0.271973	0.245382	3.078819	2.321202
312	0.257757	0.195314	0.259093	0.198785	4.282380	2.984986
400	0.265906	0.153521	0.269833	0.158972	–	–
429	0.276956	0.197973	0.281509	0.203527	3.441493	2.411608
435	0.260099	0.151096	0.266645	0.158111	6.873734	3.323427
446	0.305206	0.233673	0.306943	0.272100	2.479117	1.929448
451	0.273297	0.193117	0.276409	0.198226	4.774253	3.079460
470	0.281105	0.243139	0.284055	0.199636	4.389394	2.870175
475	0.276814	0.196772	0.278850	0.201003	5.011890	3.193580
479	0.310439	0.196746	0.312663	0.200719	5.061865	2.965935
69	0.286855	0.163151	0.292541	0.169808	8.776569	4.056630

Candidate formal endpoint IDs total 1,150 and raw IDs 1,176. All 26 suffix IDs are charged. Drafted/accepted IDs total 888/702, and target/draft continuation caller tokens 1,332/1,362. These denominators describe different runtime quantities and cannot be collapsed into one acceptance-adjusted cost. The narrowed target common-sampling measurement of 851.52 ms omits other work; it is not a full component profile. The known complete-campaign host subtotal is approximately 6,541.86 s, excluding the unknown interruption/teardown. It is not the cost of every planned request or a normalized per-task speed estimator.

S6. Historical Costs, Stops, Control, and Target-Work Budgets

S6.1. Early four-system inventory and unchanged estimators

The historical inventory uses MacBook Air M4/Metal (llama.cpp b10809), Core Ultra 9 275HX plus RTX 5070 Ti Laptop/CUDA (b10839), Core i5-13400 plus Arc B570/Vulkan0 (b10839), and Core Ultra X7 358H plus Arc B390/Vulkan0 with an OpenVINO draft. Qwen3-8B Q8_0 is the target.

Draft is Qwen3-0.6B Q8_0 except the symmetric INT4 OpenVINO adapter. Hardware, OS, backend, placement and frontend vary jointly. The inventory is not a hardware-only comparison or a common energy/latency ranking.

Table S9 preserves the closed historical outcomes. M4's representative 24Q and separately preregistered HumanEval30 have opposite measured outcomes in distinct sessions. This does not identify a workload-only cause. Early MLX precision/correctness failures are separate from the final Q8_0 Metal configuration. RTX historical CPU-to-GPU and same-GPU controls retain different integration costs. B570's formal failure triggers its stop; a later six-prompt serial text probe passes 18/18, whereas its full 30-prompt serial text comparison reaches only 60/90. Remote-interaction-contaminated diagnostic observations are excluded from the final cells. Legacy "exact" labels do not upgrade the text contract.

Table S9: Historical latency outcomes under their original contracts. Intervals are descriptive and source-specific. Text agreement is not a direct-ID certificate.

Campaign	Speed	Interval	Agreement
M4 representative 24Q	0.842	[0.809,0.877]	72/72 text
M4 HumanEval30	1.387	[1.351,1.422]	90/90 text
RTX CPU draft, HumanEval30	0.404	[0.390,0.418]	90/90 text
RTX same-GPU draft, HumanEval30	0.650	[0.629,0.673]	90/90 text
B570 serialized fallback	0.3087	[0.2927,0.3246]	60/90 text
358H CPU stateful, HumanEval30	0.733915	[0.692920,0.771090]	90/90 IDs
358H NPU stateful, HumanEval30	0.351663	[0.332595,0.368949]	90/90 IDs

Historical M4 24Q takes a within-prompt GM of three paired ratios, then resamples prompts 10,000 times with Python Random seed 20260907 and sorted draw indices 249/9749. M4 HumanEval uses the same within-prompt aggregation but 100,000 draws, seed 20260910 and indices 2500/97499. Historical RTX uses 10,000 draws, seed 20260910, indices 249/9749. B570 fallback uses 10,000 draws, seed 20260910 and indices 250/9750 in archived insertion order. Sorting tasks changes its finite-seed endpoints and is not substituted. 358H uses prompt medians of three paired ratios, 10,000 draws, NumPy default_rng seed 20260919 and .025/.975 quantiles. These policies are not replaced by late RTX's ratio-of-medians estimator.

S6.2. Target-forward proxies and energy boundaries

The historical target-only proxy is

$$V_k^{\text{micro}}(n, e) = T_{\text{target forward}}(k, n, e)/T_{\text{target decode}}(1, n).$$

It omits drafting and can omit correction/bonus, state work and complete target-round bookkeeping. Table S10 retains every context cell. Their ranges span contexts rather than uncertainty intervals. The historical RTX HumanEval/106 diagnostic repeats a same-position argmax change from 2790 to 2629 three times, with a separately instrumented same-commit margin moving from $+0.080948$ to -0.126457. It is shape-sensitive evidence in that configuration, not a unique kernel cause or a formal timing observation.

Table S10: Frozen target-forward proxy lookup. These are neither complete target-round cost nor integrated speed. Historical text agreement applies only under each original comparator.

Path	Context	V_2^{micro}
M4 Metal	512	1.022182
M4 Metal	2048	1.076126
M4 Metal	8192	1.032264
RTX batched	0	1.009030
RTX batched	256	0.983148
RTX serialized	0	2.430996
RTX serialized	256	2.137011
B570 fast	512	1.242576
B570 fast	2048	1.637295
B570 serialized	512	1.982668
B570 serialized	2048	1.991707

M4 HumanEval energy measures the Silemetry CPU+GPU+ANE package aggregate, excluding separate DRAM. Fixed-$k = 2$ package efficiency is $1.013\times$ with interval $[0.990, 1.034]$, so improvement is not established. The registered secondary $p_{\min} = 0.9$ condition has $1.041\times$ $[1.018, 1.065]$, without replacing the primary or establishing material added speed. The corrected synthetic ceiling was registered but not executed; earlier synthetic diagnostics are not a substitute.

RTX board and CPU-package sensors are distinct domains, and follow-up process windows include loading; no formal serial HumanEval energy

comparison is claimed. B570 fallback measures adapter-board efficiency $0.302\times$ $[0.285, 0.320]$, without CPU package or wall power, and its output failure confines the result to diagnosis. 358H has no qualified end-to-end energy boundary. No late runtime repair energy, common system-energy ranking or concurrent-throughput result is inferred.

S6.3. Fresh direct-ID gates and independent latency arm

Fresh RTX direct-ID v2 has 270 executions on 30 HumanEval tasks, three conditions and three repeats. Batched $k = 2$ reaches 75/90 (25/30 wholly equal prompts) and is blocked; serialized reaches 90/90 and qualifies. Mismatching first indices are /64:114, /106:65, /119:17, /128:41 and /129:21, stable across repeats. All conditions repeat stably. The initial v1 transport study retains 83 executions, 82 valid, and no qualification result: full-log UTF-8 decoding fails on a partial draft fragment after native exit zero. V2 changes observational byte transport and restarts the whole frozen schedule; it imports no v1 formal pairs. The repository's v1-path final report is a completed-v2 report copy, so its path alone does not establish v1 completion.

B570's separate fresh direct-ID campaign also completes 270 executions: fast reaches 60/90 (20/30 prompts), serialized 78/90 (26/30), and both are blocked. Historical fast 45/90 and serial 60/90 are unchanged text-path counts. An initial metadata decoding launch fails before inference with zero formal executions; authorized direct execution of the unchanged frozen runner produces the full campaign. No formal speed is inferred from operational elapsed fields. Neither new gate retroactively qualifies old timings.

The independent RTX matched-latency arm has 180 fresh timing observations and 90/90 fresh complete-ID agreements on 30 tasks. Serialized speed is $0.645816\times$, interval $[0.629802, 0.659884]$, zero of 30 faster. Condition order uses seed 2026093001; the 10,000 prompt-bootstrap draws use seed 2026093002 and linear percentiles. Native QPC timing starts after synchronized target/draft prefill and covers per-request sampler/state work, drafting, verification, acceptance/correction/bonus, rollback, synchronization, logging/response and slot release. Load and teardown are excluded. Chrono shares the host oscillator; it is not an independent physical clock. The empty-hook probe bounds clock-call overhead, not all logging overhead. The fresh frontend is compiled with Clang 21.1.0 over retained CUDA libraries; OS build 26300 differs from the earlier direct-ID OS26200. These facts preclude synthetic pairing with prior qualification.

S6.4. V3 stop, V4 recovery, and protocol amendment

V3 plans 576 formal condition/repetition rows over 24 tasks. It delivers 126 valid prefix rows: 72 pre-live and 54 cost, with two failures and 448 unreached rows. Native HumanEval/59 emits 128 committed IDs ending at EOG and reports both limit and EOS. The frozen parser accepts only one marker and stops. Exit zero is not sufficient to make capture valid. A-live receives zero formal runs; complete formal accuracy, speed and interval are unavailable. Engineering qualification's 400 runs and parser-negative fixtures do not repair the missing formal endpoint branch. Old failures remain sealed.

V4 fixes that observational endpoint handling under a new freeze, accepting the ordered complete EOS-at-cap case while rejecting contradictory, reversed, duplicate or missing markers. Its 19-task panel initially comprises the remaining project-unused HumanEval inputs; subsequent reuse is exposed. It retains 228 pre-live predictor records and adds 57 native live and 171 fresh timing observations: 456 complete formal condition rows. Prediction closure is described in Section S2. The cost arm uses one timer executable and balanced order, each condition appearing in each order position 19 times. Batched qualification remains 51/57 and has no admissible speed/interval. Global serialization qualifies 57/57 but yields $0.677351\times$ $[0.661653, 0.691812]$, zero of 19 faster. Median task absolute times are 1.607 s target and 2.332 s serial; their ratio is not the paired GM. The interval uses 10,000 paired-row draws, seed 2026100204 and linear quantiles.

The V4 numerical sources and assets remain frozen, but a fan-curve change triggers the original strict host-hash stop before live outcomes. Authorized continuation preserves the original `PROVENANCE_INVALID` records, repeated predictor checks and the amendment. Fan/lighting differences become permitted; non-fan material settings remain strict. Predictor data are reused, not relabeled as a fresh untouched cohort. Fan controls stay fixed during the fresh timing arm, with snapshots retained, but can still affect latency. This is disclosed amendment completion, not a claim that the original no-deviation host policy passed. Instrumented post-prefill-to-slot-release time is distinct from the later server generation field and cannot form a common optimization curve.

S6.5. RTX efficiency screening

The early FA-on/UB512 four-probe pilot qualifies small-model $k = 2$ and $k = 4$, with request/generation speeds $1.277\times$ /$1.324\times$ and $1.492\times$ /$1.572\times$. Each FA/UB setting has its own ordinary reference; global UB changes several

phases. Expansion to HumanEval/0–7 preserves every failure: final small-model $k = 2$ qualifies 18/24, fragment drafting 21/24 and the mixed strategy 15/24. A mixed result producing 18 rather than the reference 87 IDs is invalid as a speed claim. $k = 2$ repairs /5 but introduces /2; /6 remains divergent. The campaign has 300 formal requests and 48 excluded warmups. Synthetic favorable cases do not support the current headline.

S6.6. M4 formal control and oracle opportunity

The M4 Guard measures cumulative charged generation time per committed ID, including seed/correction/bonus and excluding rejected proposals. It starts on Always speculation and can fall back permanently after a completed round, reconciling invalid target/draft suffixes first. Calibration freezes target cost per ID and selects `E8_M000` from a nine-candidate grid with rounds $\{2, 4, 8\}$ and margins $\{0, 0.05, 0.10\}$. Exactness and calibration regret select the rule; ties use frozen overhead/exploration/margin/ID rules. Disjoint calibration, qualification and held-out task IDs prevent held-out retuning. Online decisions receive no future length, task family, request-specific target latency or oracle answer.

The registered post-prefill clock includes generation initialization, seed sampling, draft/target work, synchronization, reconciliation, control and transition. Load, sampler/speculator setup, initial prefill and the speculative begin hook precede it. This is not whole HTTP request latency. The formal campaign has 60 held-out tasks, 30 each HumanEval and Spec-Bench, three conditions and three repeats. Always and Guard each qualify 180/180 IDs. Table S11 gives their original overall and family estimates.

Table S11: M4 formal post-prefill baseline/candidate speeds from three-repeat condition medians. Family strata are descriptive.

Panel	Tasks	Target/Always	Target/Guard
Overall	60	1.346894	1.311387
HumanEval	30	1.570011	1.554690
Spec-Bench	30	1.155485	1.106160

The primary 10,000-draw bootstrap uses Python Random seed 20260924 in held-out order, with one continued stream for target/Guard, target/Always and Guard/oracle. The paired Guard/Always secondary replay starts a fresh stream of the same seed and uses linear percentiles; it gives latency 1.027076

[1.012956, 1.043559], with Guard faster on 24 and Always on 36 tasks. It is post-study, not the original preregistered endpoint. A separate family-stratified sensitivity uses seed 2026100101, 10,000 draws and HumanEval-then-Spec-Bench order, giving [1.0135, 1.0423]; it leaves the original intervals unchanged.

The per-task two-fixed-mode oracle improves the measured interval by factor 1.0006919671 relative to Always. The implied latency reduction is $1 - 1/1.0006919671 = 0.0006914886$, approximately 0.069149%. Only Spec-Bench/559 and /95 favor target-only: target times are 11.9085 and 10.9203 s; Always/target latencies 1.0270 and 1.0150; Guard/target 1.0215 and 1.0029. This finite hindsight opportunity is not a bound on all intra-request policies.

A loop-variable shadowing defect records a build-manifest digest in the intended calibration-freeze field. Recovery verifies the actual freeze, source/model/calibration/policy hashes and formal-start links, retaining error markers and the original pipeline `STOPPED` metadata. It does not retune the policy or mutate measurements. Recorded Git order is not a trusted external timestamp. This resolved metadata defect is not described as an unfinished formal experiment.

S6.7. 358H state interface, threshold gate, and conditional budgets

The completed 358H stateful runner supersedes an empty blocked-preflight scaffold. It keeps the target's low-level context between rounds and charges validation, rejected-suffix rollback, correction/bonus and synchronization within complete target work. Every OpenVINO proposal call receives the full committed prefix. The synchronous region combines replay/reconstruction and proposal, with no separate measurements of those terms.

A prior proxy threshold study passes 18 direct-ID sanity pairs and admits borderline context points, using a favorable zero residual T_x, fixed progress assumptions and seed 20260919, 10,000 empirical draws. It predicts eligibility, not achieved integrated speed. CPU central/optimistic speeds at contexts512/2048/8192 are 0.903680/1.355520, 0.884843/1.327264 and 0.755946/1.133919. NPU values are 0.905553/1.358329, 0.903804/1.355706 and 0.839618/1.259427. The subsequent actual stateful CPU/NPU speeds in Table S9 are lower and qualified, with zero of 30 prompts faster on either path.

Table S12 retains measured progress, components and distinct ceiling estimators. Both paths commit 11,520 IDs, over 4,677 CPU and 4,686 NPU rounds. Arithmetic-total accounting gives normalized $(D, R, X) =$

$(0.4304, 2.9767, 0.00124)$ CPU and $(4.1251, 2.9435, 0.00130)$ NPU. These values reproduce total-time ratios, not prompt GMs. Target-only ceilings assume unchanged baseline, target work and progress with all other nonnegative cost removed.

Table S12: 358H measured accounting and conditional zero-other-work ceilings. Percentages describe the saved synchronous regions; prompt and pooled estimators remain distinct.

Quantity	CPU draft	NPU draft
Committed IDs per round	2.463117	2.458387
Replay plus proposal share (%)	12.6273	58.3474
Complete target-work share (%)	87.3365	41.6342
Residual share (%)	0.0363	0.0184
Actual pooled-total speed	0.7227	0.3477
Pooled-total ceiling	0.8274529834	0.8351858524
Prompt-estimator ceiling	0.8372192092	0.8448401086
Necessary uniform target cut, prompt (%)	16.278079	15.515989
Necessary uniform target cut, pooled (%)	17.254702	16.481415

No prompt-median ceiling reaches parity; maximum prompt ceilings are 0.9736196764 CPU and 0.9999089575 NPU. One NPU repetition has $B/T = 1.000163695$, so not every individual run-level ceiling is below one. The budgets are necessary arithmetic reductions under fixed progress, not achieved optimizations or proof against NPUs. A persistent-state/wider-target candidate stops at qualification: CPU restore/reconcile correctness is unavailable under the tested representation, and the frozen dynamic-shape NPU IR lacks the required state-capable path. Same-prefix target-only and proposal-replay diagnostics identify greedy divergence but not a unique rollback bug. No formal runtime performance or energy result exists for that failed candidate; the completed earlier Phase-3 costs remain valid under their own contract.

S7. Arithmetic Reproduction, Provenance, and Remaining Access Limits

The author S2 package provides editable LaTeX tables/equations, separate vector PDF/SVG figures, bibliography, Highlights, CPU reconstruction and build code, selected authenticated report/implementation records, claim–evidence mapping and the complete experiment-destination ledger. It includes

the retained Elsevier class and numeric bibliography style. The two document entry points are `main.tex` and `supplement.tex`. Compile either directly in Overleaf; the documented local build regenerates tables and figures from the frozen exports before compilation.

The inherited 26-file evidence binding and 16 historical source copies remain byte-for-byte unchanged. Reproduction checks reconstruct the RTX 384-row matrix, qualification flags and response-ID digest bindings, medians, all 12 comparisons, descriptive intervals, task regressions, round reversal and the ratio identities. They independently reconstruct B570's 119 formal rows, 15 versus 14 complete task pairs, all three cost timers and paid suffix counts. The phase task booleans, natural-state matrices, fixed-UB task-ratio aggregation and M4 medians/oracle are also checked. Counts inferred from compact exports are distinct from a renewed audit of every original raw payload.

The original runtime reports contain stronger original-host audits, retained external-asset hashes and large content-addressed archives. This reconstruction reads those reports and source, authenticates the copied Git blobs, and recomputes included exports. It does not reconstruct every private archive, locally rehash omitted model/driver binaries, verify independent absolute execution time or run new GPU experiments. Numerical reproduction of a summary or task-ratio export must not be called a full raw-data audit. Missing B570 payloads and unknown suffix comparisons remain unknown.

Clean source-package extraction and rebuild check document text, citations, embedded fonts and build diagnostics. Final PDFs are rendered and inspected page by page, including figure labels and table continuations. Numerical and build QA records state actual executed checks; visual QA records identify the inspected final pages. These are author-side validation activities, not external peer review or commercial plagiarism checking. The draft PR is a review entry point rather than journal submission.

HumanEval, MBPP and Spec-Bench source descriptions are cited in the main manuscript [1, 2, 3]; software is cited directly [4, 5]. Web/preprint versions and access dates are recorded in the bibliography and reference audit. Authors must finalize authorship, affiliations, CRediT, funding, competing interests, permissions, permanent data/software access and the AI-assistance statement before submission. A second model family, external method implementation comparisons, stochastic sampling and concurrent throughput are outside the completed evidence scope; no future outcomes are supplied here.

References

[1] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. Ponde de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. Petroski Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. Hebgen Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, W. Zaremba, Evaluating large language models trained on code, arXiv preprint arXiv:2107.03374v2, accessed 9 October 2026 (2021). doi:10.48550/arXiv.2107.03374.
URL https://arxiv.org/abs/2107.03374v2

[2] J. Austin, A. Odena, M. Nye, M. Bosma, H. Michalewski, D. Dohan, E. Jiang, C. Cai, M. Terry, Q. Le, C. Sutton, Program synthesis with large language models, arXiv preprint arXiv:2108.07732v1, accessed 9 October 2026 (2021). doi:10.48550/arXiv.2108.07732.
URL https://arxiv.org/abs/2108.07732v1

[3] H. Xia, Z. Yang, Q. Dong, P. Wang, Y. Li, T. Ge, T. Liu, W. Li, Z. Sui, Unlocking efficiency in large language model inference: A comprehensive survey of speculative decoding, in: Findings of the Association for Computational Linguistics: ACL 2024, Association for Computational Linguistics, 2024, pp. 7655–7671, accessed 9 October 2026. doi:10.18653/v1/2024.findings-acl.456.
URL https://aclanthology.org/2024.findings-acl.456/

[4] ggml-org llama.cpp contributors, llama.cpp [software], Releases b10809 and b10839; private study patches documented in S2, b10809 released 4 September 2026; b10839 released 7 September 2026. Accessed 9 October 2026 (2026).
URL https://github.com/ggml-org/llama.cpp/releases/tag/b10839

[5] A. Yang, et al., Qwen3 technical report, arXiv preprint arXiv:2505.09388v1, accessed 9 October 2026 (2025). doi:10.48550/arXiv.2505.09388.
URL https://arxiv.org/abs/2505.09388v1